

POLITECNICO
MILANO 1863

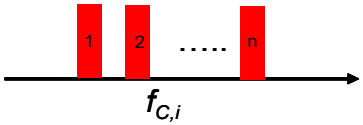
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

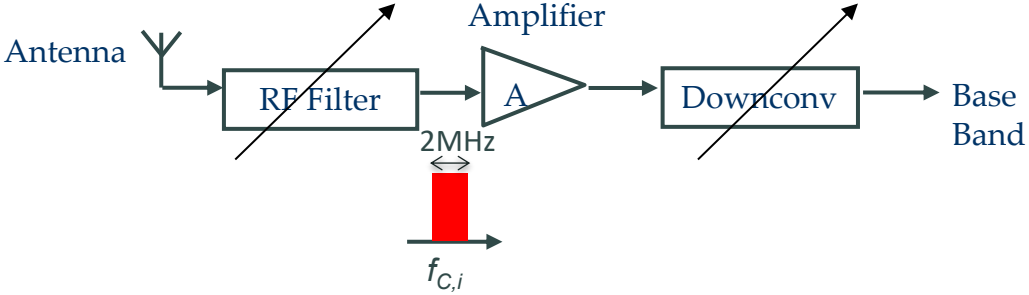
RF Systems

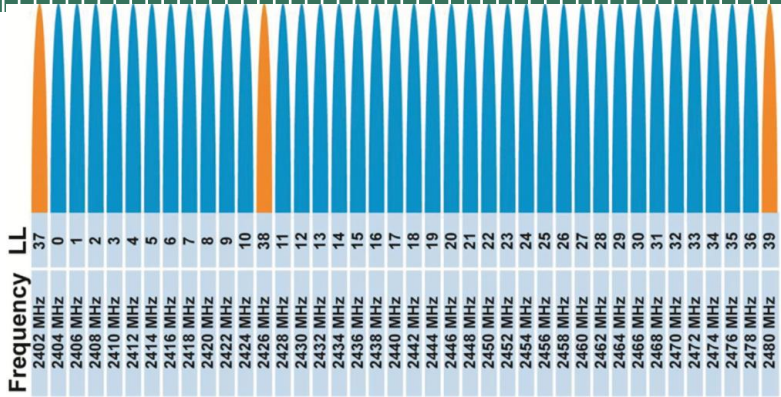
RX Architectures:
Single-Conversion Receivers & Components
Non-Linear Phenomena (spurious, intermodulation, blockers)
Other architectures (double-conversion, direct conversion)
Other components (image-reject mixer, hybrid splitter)

Direct Detection of Radio Signals



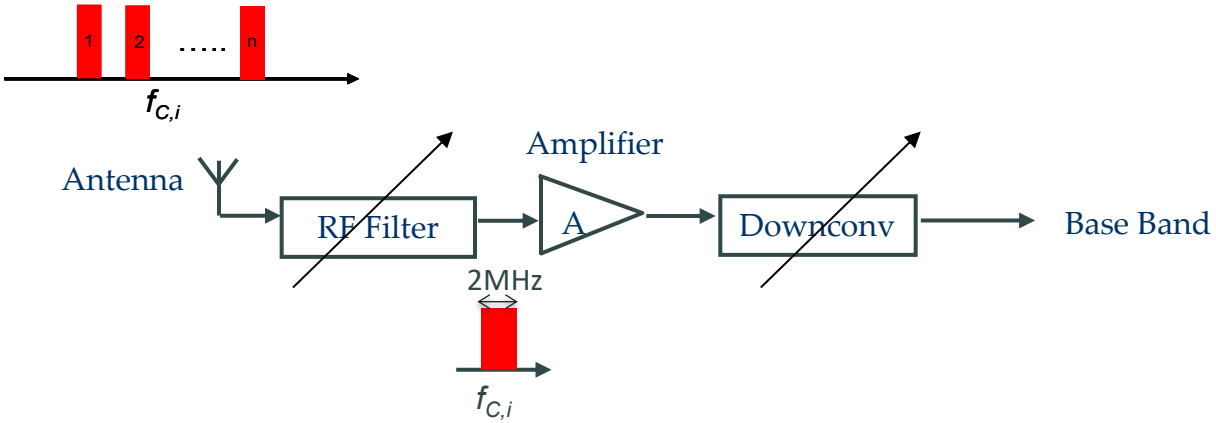
A RF system must receive one of several closeby Bluetooth channels (2MHz each, from 2400 to 2480MHz)





Why is this architecture inconvenient?

Direct Detection of Radio Signals



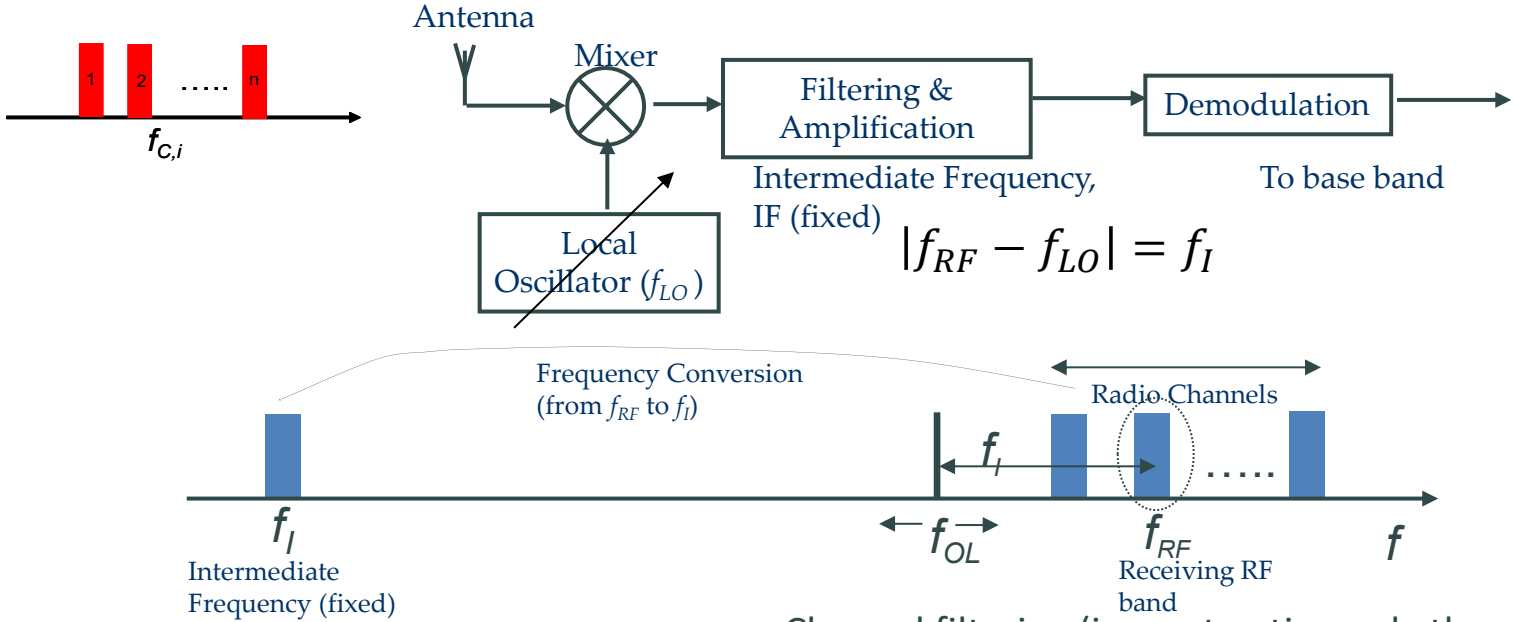
Not convenient:

- RF filter must be very selective and tunable
- Amplifier operating at RF with high gain and low noise required
- Poor performances even with expensive components

3 main practical receiver architectures:

- Single Conversion
- Double Conversion
- Direct Conversion (or zero-IF)

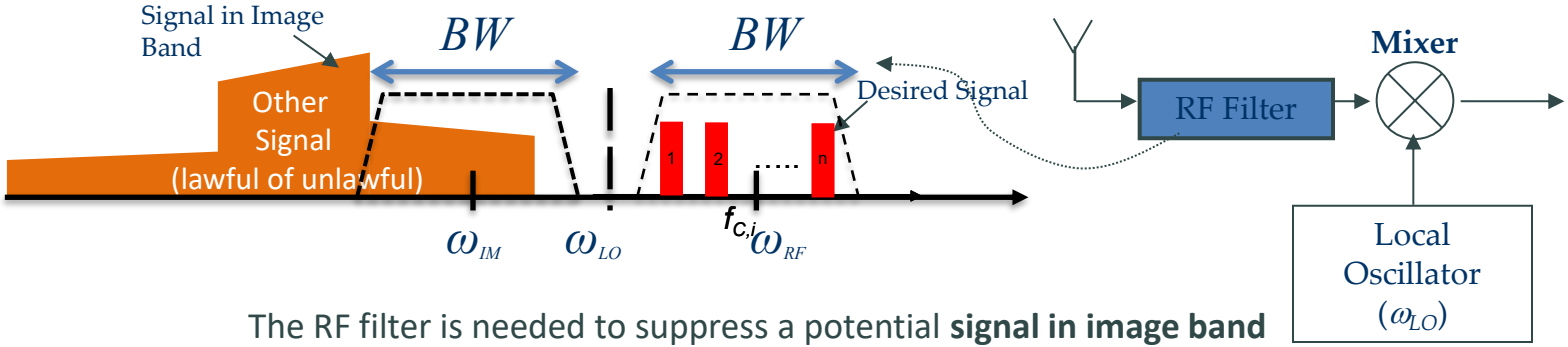
Conversion Receiver (Concept)



The tuning of RF channels is obtained by varying the frequency f_{LO} of the local oscillator!

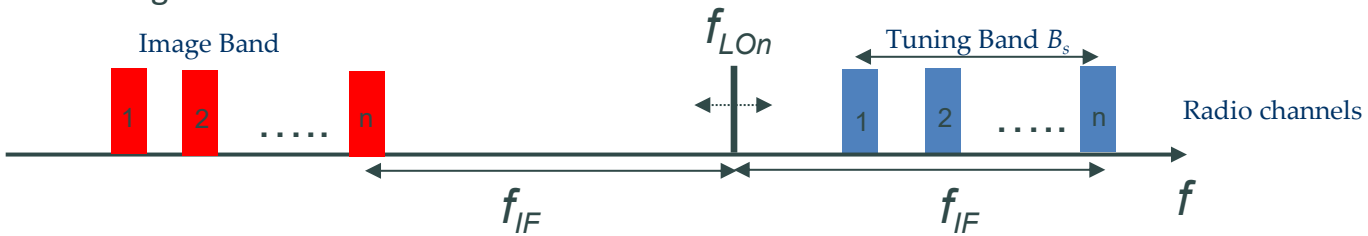
Channel filtering (i.e., extracting only the required narrow-band channel among many) can be done at IF
→ IF filter can work as channel filter

Image Frequency Problem (already seen)

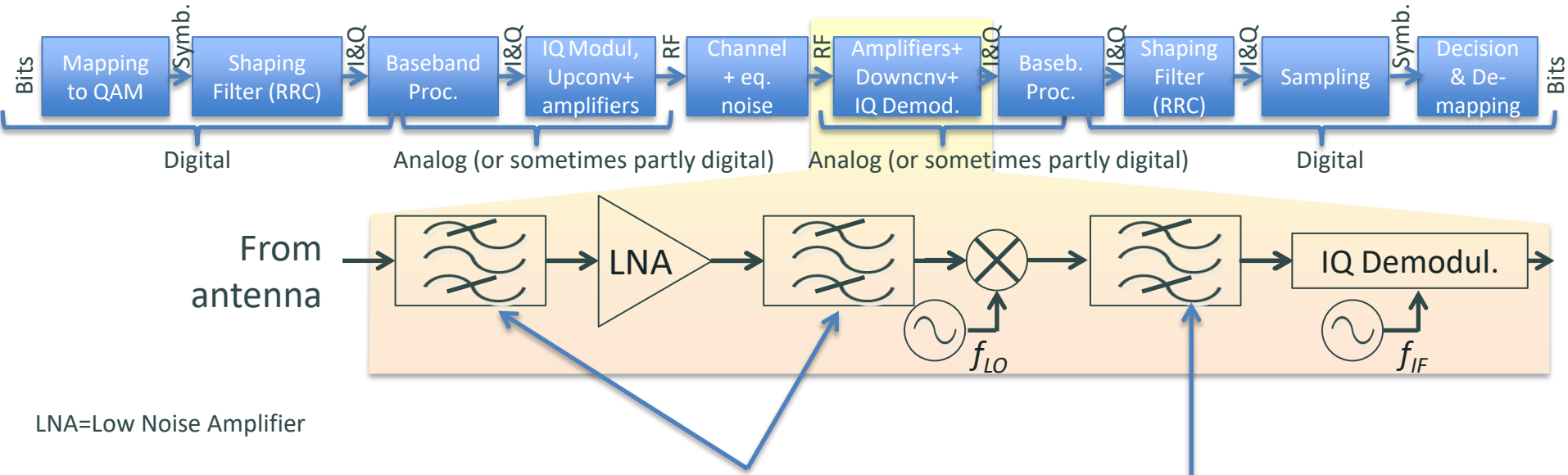


The RF filter is needed to suppress a potential **signal in image band**

The tuning band B_s is separated from the image frequency of twice the value of the intermediate frequency. The RF filter (with passband equal to the tuning band) must suppress the image band



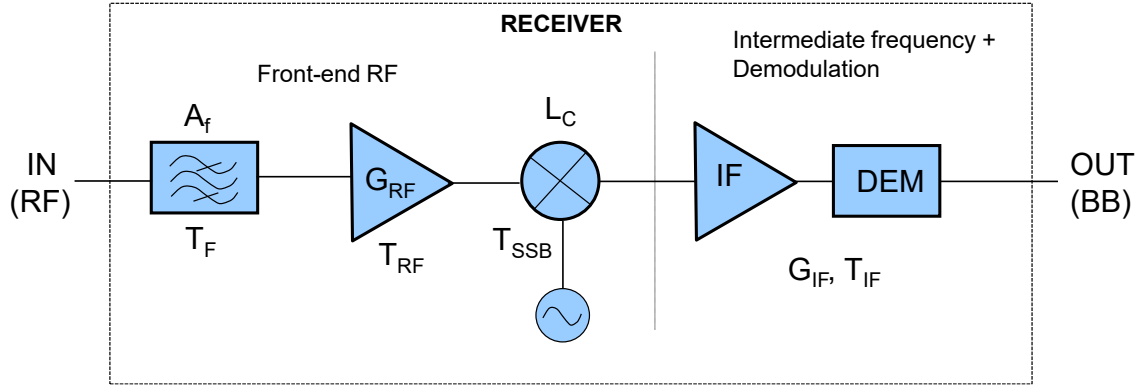
Single-Conversion Receiver

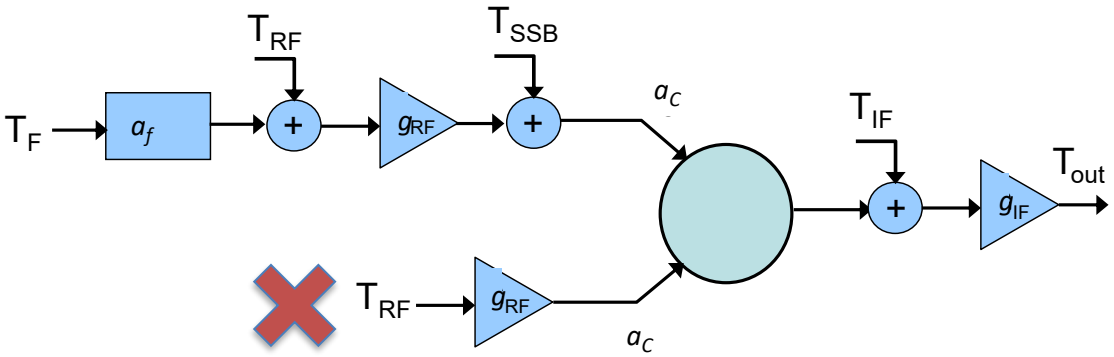


LNA=Low Noise Amplifier

The first filters in RF section are microwave filters (distributed parameters).
The IF filter is usually a lumped-element circuit (i.e. capacitors, inductors…)
The use of the LNA and second RF filter is not strictly necessary (see later)

Recap: Evaluation of the receiver Noise with infinite rej.





If the IM filter has ∞ rejection in IM band (As=∞dB), what comes before is suppressed!

Recap: Evaluation of the receiver Noise

The equivalent noise temperature of a receiver is obtained by dividing the noise temperature at output for the overall gain between BB and antenna:

$$T_{REC} = T_{out} \frac{a_C a_f}{g_{RF} \cdot g_{IF}}$$

From the scheme in the previous slide, assuming $A_f(f_{IM}) \approx \infty$, we obtain:

$$T_{out} = g_{IF} \cdot \left[\frac{T_{RF} \cdot g_{RF}}{a_C} + \frac{\left[\frac{T_F}{a_f} + T_{RF} \right] g_{RF} + T_{SSB}}{a_C} + T_{IF} \right]$$

$$\begin{aligned} T_{REC} &= T_{RF} \cdot a_f + \left[T_F + a_f T_{RF} \right] + \frac{a_f T_{SSB}}{g_{RF}} + \frac{a_C a_f T_{IF}}{g_{RF}} \\ &= T_F + 2 a_f T_{RF} + \frac{a_f (T_{SSB} + a_c T_{IF})}{g_{RF}} \end{aligned}$$

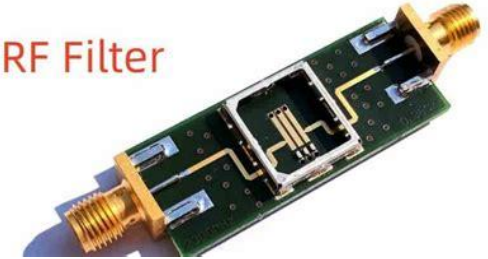
Filters play an important role: image band rejection, undesired signal suppression and …
they limit out-of-band noise
But practical filters are far from ideal!

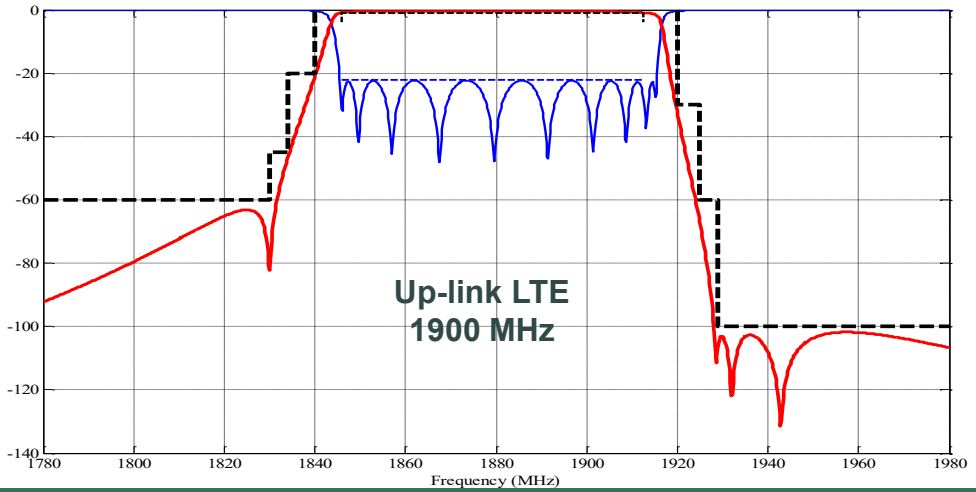
Characteristic Parameters of RF Filter

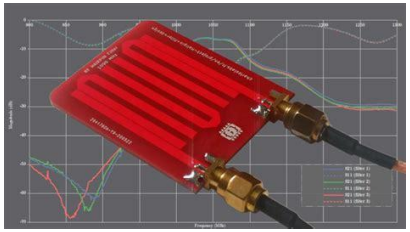
Selectivity (defined by means of the Transducer Gain/Attenuation Mask)

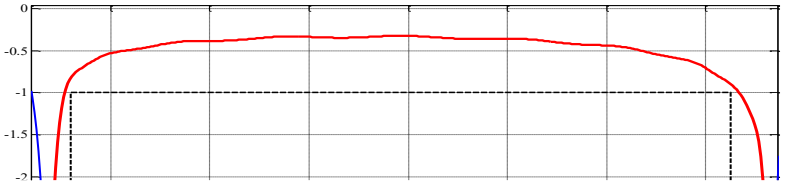
Insertion loss in passband (dB) (worsens the receiver S/N!)

Physical parameters (dimensions, weight, etc)







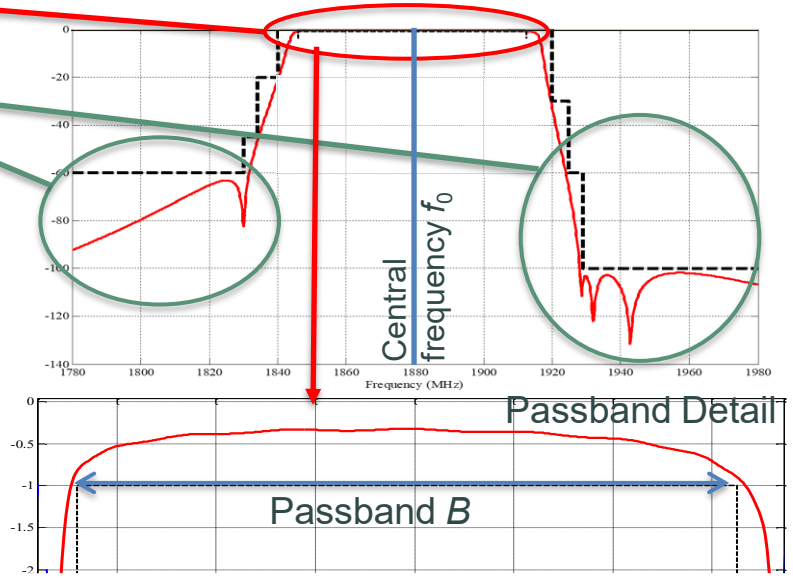


Passband Attenuation

Considerations on microwave filters

- The passband loss is due to dissipation
  → noise is generated
- Out of band rejection is produced by reflection
  → no noise produced
- Microwave filters are constituted by *N* cavity resonators coupled to each other. Higher num. of cavities
  → Higher selectivity, but higher passband losses
- For a given *N*, passband loss depends on the unloaded quality factor of cavities
- Larger cavities/components→Higher Q → Lower loss
- For a given Q, the loss increases with the decrease of B/f_0 ratio: narrow band filters → Higher loss
- Rejection in stopband: increases with order of filter (and finite transm zeros). **Rejection over a stopband is finite** (e.g. 60dB…)

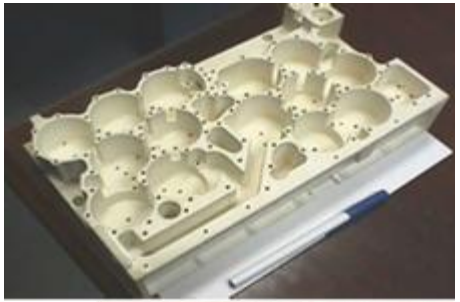
Gain in dB ($Gain_{dB}$=-$Attenuation_{dB}$)
(Gain<0dB since this is a passive component)

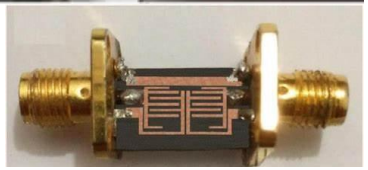


E.g. Passband gain=-1dB → Passband attenuation=1dB (dissipative attenuator)

Microwave filters technology

- The highest Q can be realized with air-filled 3D cavities with dimensions related to the wavelength at the center frequency (Q values ranging from 1000 to over 10000)
- Planar technology is widely used today in microwave circuits (also in the RF front-end). The Q obtainable in this case is relatively low: 100-300
- For (portable) devices operating below 2 GHz, special integrated technologies are available (SAW, FBAR) allowing a very compact implementation at expense of high losses (>2-3 dB)

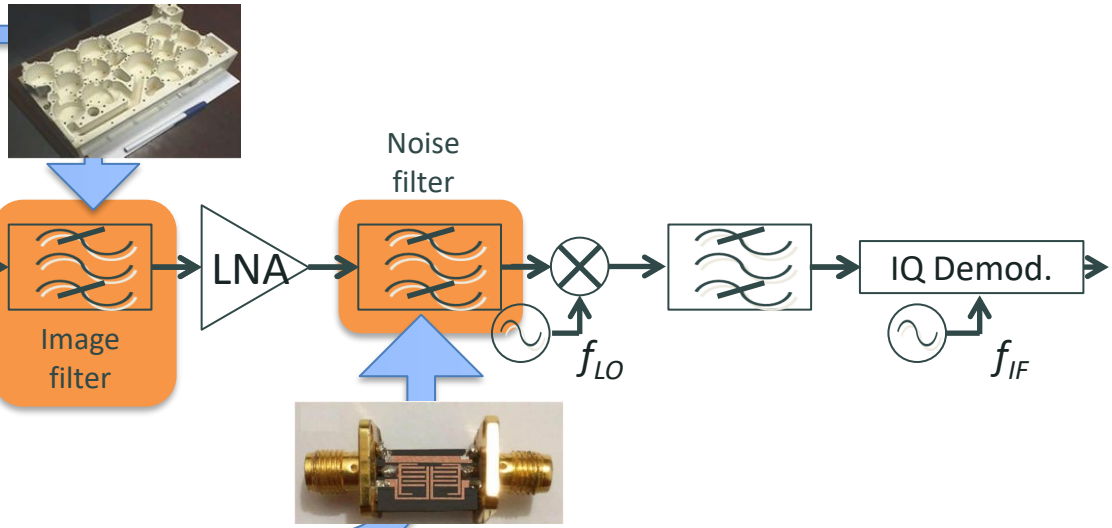




Are RF filters necessary?

Image Filter (IM Flt):

- reject the image band
- remove strong interferers which would saturate the LNA
- degrades the overall noise figure of the receiver its passband loss
  → It must be low loss
  → high Q (e.g. 3D technologies, such as waveguide)



Noise Filter :

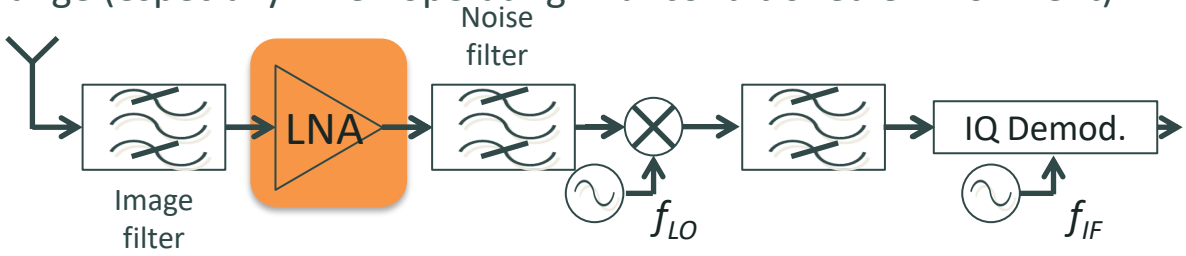
- rejecting wideband noise by LNA. Not necessary if NF is very small
- Filter 2 is after LNA so its contribution to overall NF is divided by G_{LNA}
  →Higher attenuation is tolerated (lower Q of the cavities, planar technology possible)

Key Figures of LNA

LNA's purpose: increase the power level as required by the mixer for proper operation

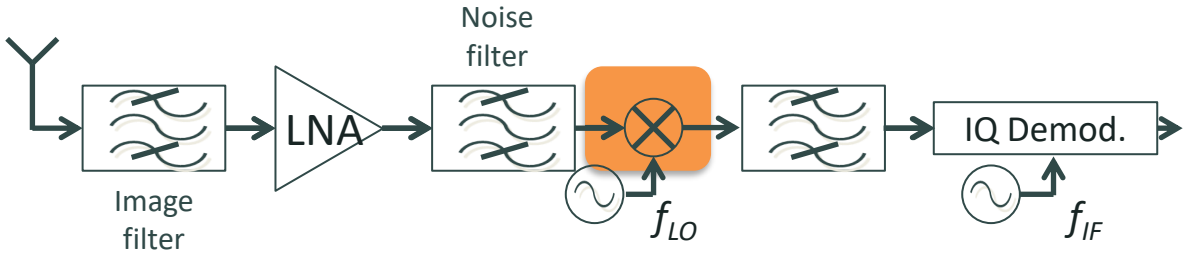
It introduces additional noise, but the overall S/N ratio of the receiver can be improved!

- Transducer Power Gain (in dB)
- Noise Figure (degradation of S/N ratio from input to output)
- Maximum power level at output (linearity)
- Input and output matching
- Biasing (Voltage and Current. DC power required)
- Temperature range (especially when operating in unconditioned environment)



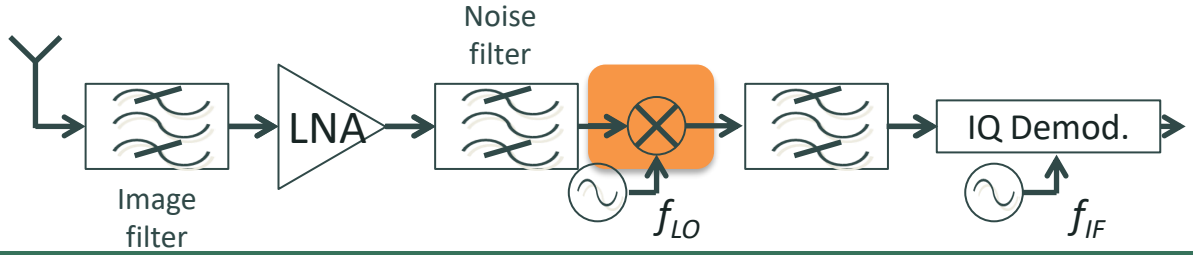
Mixer

- Translate the RF signal to intermediate frequency (IF)
- At microwave frequencies it is implemented via analogical multiplication or exploiting the non linearity of particular components (diodes or transistors)
- It affects the receiver performances: adds noise (T_{SSB}) and degrades linearity (IIP3)

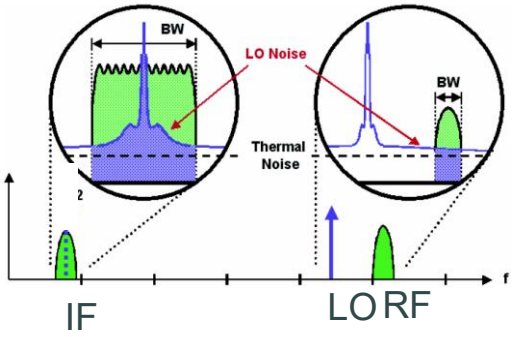


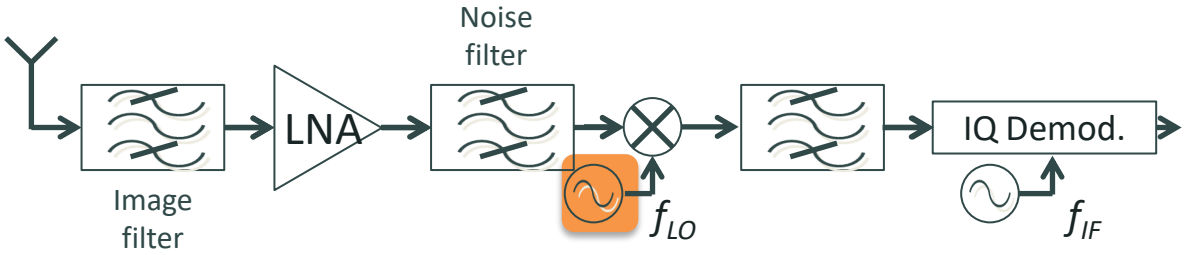
Microwave Mixers Parameters

- Conversion Loss (dB) (5-15 dB)
- NF_{SSB} (5-15 dB)
- Linearity (Input Intercept point IP3: 0-10 dBm)
- Local Oscillator Power (0-10 dBm)
- Spurious signals at output
- Matching at the ports (>15 dB)
- Isolation LO-IF and RF-IF (depends on the actual implementation)
- Phase noise of LO (e.g. giving carrier-to-noise>100 dBc (dB from carrier) @Δf≈100KHz-1MHz)



Local Oscillator





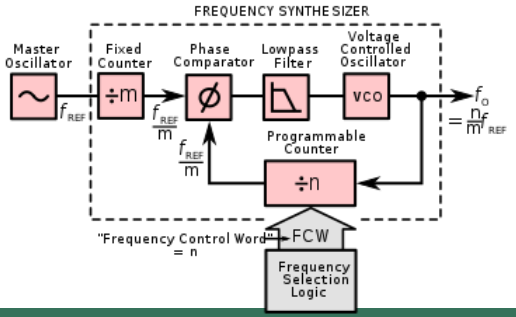
The LO is not an ideal single tone
Fluctuations of the instantaneous phase produce a broadening of the line called "**phase noise**"

- The noise spectrum around LO is transferred at IF around the translated signal degrading the SNR
- Oscillators used in receivers must satisfy strict requirements concerning the phase noise (characterized by C/N ratio at a specified distance from f_0)
- The degrading effects of phase noise can be limited by increasing the gain of the stages before

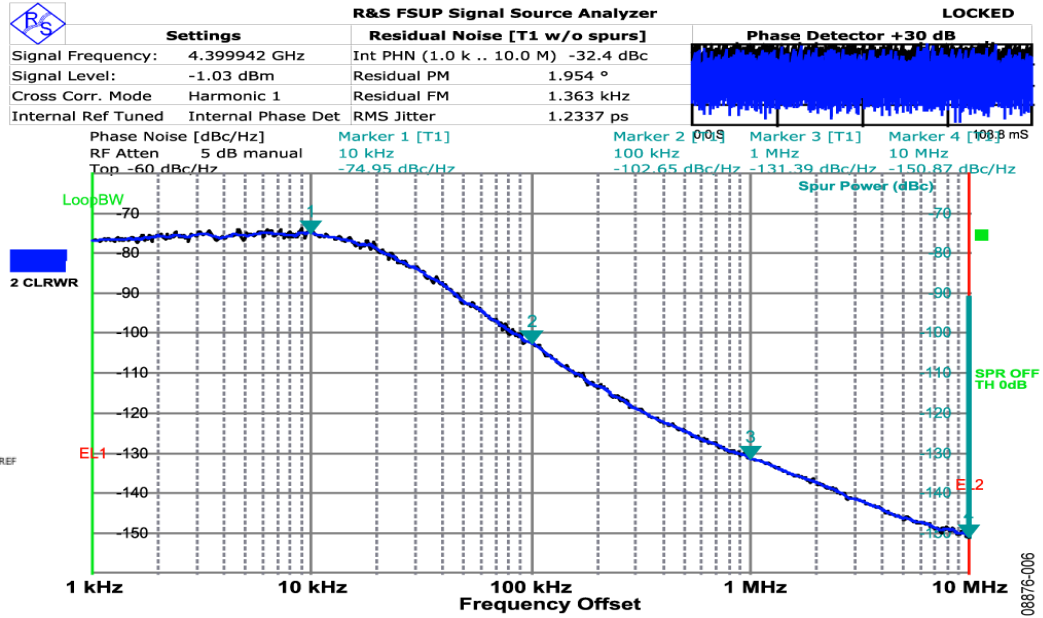
Phase Noise Measurement

Tunable oscillators:

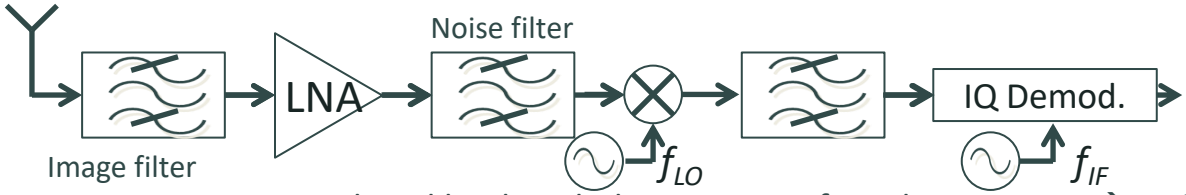
- Free-running analog controlled: Voltage-Controlled Oscillator (VCO). Long-term stability is very bad (temperature, aging…): seldom used
- Synthesized: locked at a lower frequency; possibly then multiplied (by means of non-linearities)



Spectrum of a VCO operating at 4.4 GHz

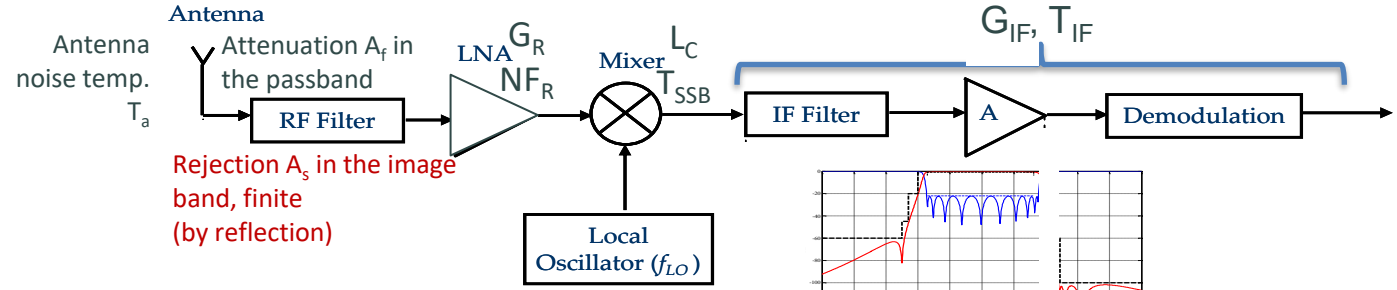


Key Figures of a Receiver

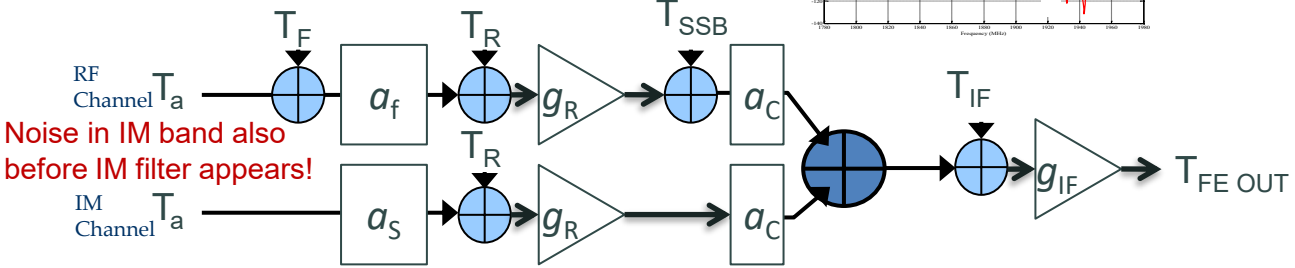


- **Equivalent noise temperature**: noise produced by the whole receiver, referred to its input → **Noise figure of receiver**
- **System noise temperature:** total equivalent noise (receiver+antenna+other components in between) referred to input
- **Sensitivity**: ability to detect a weak signal: level of the input signal (dBm) giving a specified quality (SNR or BER or EVM…)
- **Selectivity**: ability to reject the unwanted signals on the adjacent frequency channels (e.g. >70 dB)
- **Spurious rejection**: ability to reject signals outside the receiving band (given by filters and IF, OL frequencies)
- **Intermodulation rejection**: ability to withstand two or more strong interfering signals at input, generating intermodulation products converted to IF (it depends primarily on the linearity of the receiver)
- **Frequency stability**: ability to remain tuned at the correct Rx frequency in long term (drift) and short term (jitter, phase noise)
- **LO Isolation**: leakage of LO through the mixer, which may radiate or saturate following stages
- **Input IP3:** overall equivalent input 3rd-order intercept point

Example: System Noise Temperature with non-ideal filter



T_a=150K
A_f=1dB, **A_S=10dB (IM reject.)**
G_R=15dB, NF_R=2dB
L_C=5dB, T_{SSB}=400K
G_{IF}=20dB, T_{IF}=500K



Noise in IM band also before IM filter appears!

Convert to linear scale and temperatures:
$a_f=10^{(1/10)}=1.26$; $T_f=290K\ (A_f-1)=75.1K$
$a_S=10^{(10/10)}=10$; $NF_R=10^{(2/10)}=1.58$
$g_R=10^{(15/10)}=31.6$ $T_R=290(1.58-1)=168.2K$
$a_C=10^{(5/10)}=3.16$ $g_{IF}=10^{(20/10)}=100$

Total out temp.: $T_{FE\,OUT}=g_{IF}T_{IF}+\frac{g_{IF}}{a_C}T_{SSB}+\frac{g_{IF}}{a_C}g_RT_R+\frac{g_{IF}}{a_C}\frac{g_R}{a_f}(T_F+T_a)+\frac{g_{IF}}{a_C}\frac{g_R}{a_S}T_a+\frac{g_{IF}}{a_C}g_RT_R$ ⇒ $T_{SYS}=T_{FE\,out}\frac{a_fa_C}{g_{IF}g_R}=746.7K$

Or directly System Noise Temp. (equiv. input noise temp.): $T_{SYS}=T_a+T_f+a_fT_R+\frac{a_f}{g_R}T_{SSB}+\frac{a_f}{g_R}a_CT_{IF}+T_R\frac{g_R}{a_C}\frac{a_f}{g_R}a_C+T_a\frac{1}{a_S}\frac{g_R}{a_C}\frac{a_f}{g_R}a_C$

=150K+75.1K+211.9K+15.9K +63K +211.9K +18.9K=746.7K

Receiver Sensitivity

In an analog receiver:

The Sensitivity (S) is the minimum signal power at the receiver input which guarantees a desired equivalent SNR at the input:

$$SNR = \frac{S}{kT_{sys}B}$$

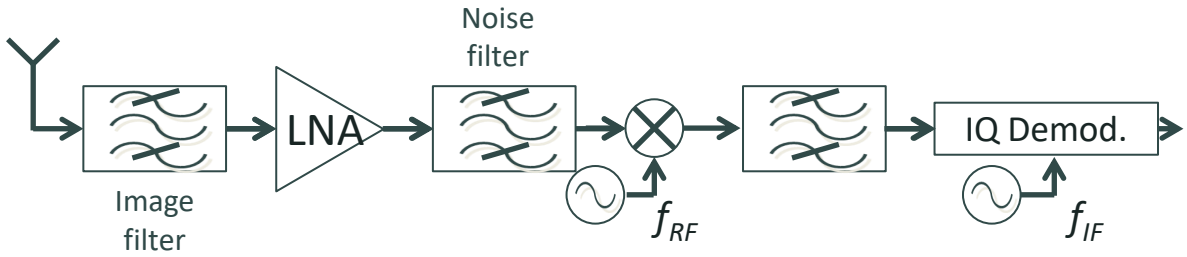
T_{eq}=System noise temperature (at input)
B = Signal band

In a digital receiver:

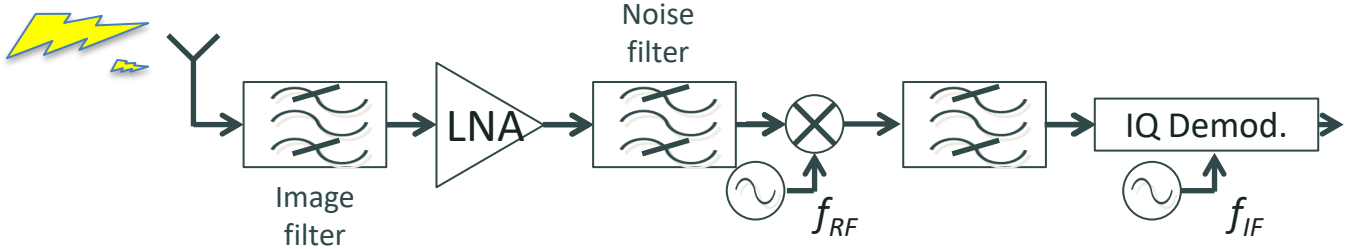
The Sensitivity (S) is the minimum signal power at the receiver input for a specified BER
The required BER determines the needed E_b/N_0, which can be translated to the required SNR

Key Figures of a Receiver

The receiver is designed to operate at its minimum power level (sensitivity) still guaranteeing the required BER.
For what practical reasons should a receiver instead observe a much higher received power and must still be able to operate?



Dynamic Range



It represents the range of power allowed at the receiver input

The minimum power is defined by the sensitivity (S)

The maximum power is conventionally assumed as the mean power of an input 2-tone signal producing a IM3 power (referred to the input) equal to the receiver sensitivity (S):

P of each 3rd order intermod at output: $P_{2\omega_2-\omega_{1\,dBm}} \approx 3P_{\omega 1_{dBm}} - 2IIP_{3_{dBm}} - 2G_{dB}$

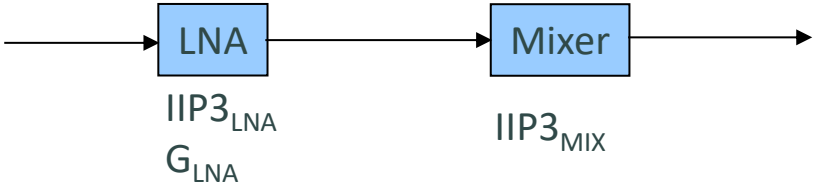
Equivalent P of 3rd order intermod at input: $P_{int_{dBm},input} = 3P_{m_{dBm}} - 2IIP_{3_{dBm}} - 6_{dB} - 3G_{dB} = 3Pmax_{dBm} - 2IIP_{3_{dBm}} - 6_{dB}$

Set $P_{int\ dBm,input}=S_{dBm}$, obtain $P_{max\ dBm}$ and finally compute DR_{dB}:

$$DR_{dB} = Pmax - S = \frac{2}{3}\left(IIP_{3,dBm} + 3 - S_{dBm}\right)$$

IIP3: 3rd order Intercept power at input of the receiver

Overall IIP3



The 3rd order intercept at input of each stage (IIP3) is equal to the intercept at output (IP3) divided by the gain

$$\left(\frac{1}{iip_3}\right)^2 = \left(\frac{1}{iip_{3,LNA}}\right)^2 + \left(\frac{g_{LNA}}{iip_{3,mix}}\right)^2$$

Dominated by last stage!

Note: all the quantities here are in linear scale (not dB!)

For a chain with *n* stages

Realistic version (power sum of IM products)

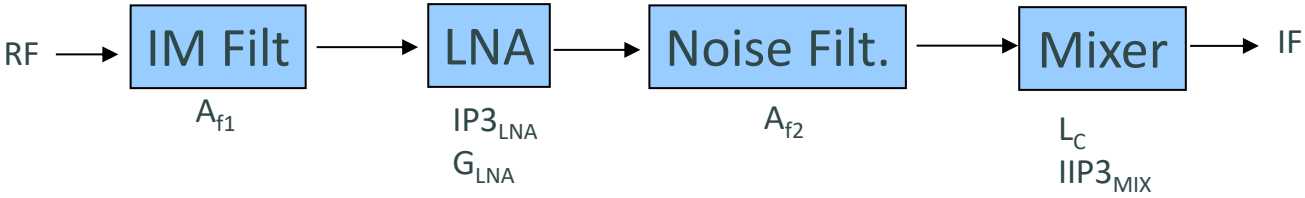
$$\left(\frac{1}{iip3}\right)^2 = \left(\frac{1}{iip3_1}\right)^2 + \left(\frac{g_1}{iip3_2}\right)^2 + \left(\frac{g_1 g_2}{iip3_3}\right)^2 + \ldots \left(\frac{g_1 g_2 \cdots g_{n-1}}{iip3_n}\right)^2$$

Worst-case version (voltage sum of IM products)

$$\left(\frac{1}{iip3}\right) = \left(\frac{1}{iip3_1}\right) + \left(\frac{g_1}{iip3_2}\right) + \left(\frac{g_1 g_2}{iip3_3}\right) + \left(\frac{g_1 g_2 g_3}{iip3_4}\right)$$

Example of equivalent IIP3

$$\left(\frac{1}{iip3}\right)^n = \left(\frac{1}{iip3_1}\right)^n + \left(\frac{g_1}{iip3_2}\right)^n + \left(\frac{g_1 g_2}{iip3_3}\right)^n + \left(\frac{g_1 g_2 g_3}{iip3_4}\right)^n \quad n = 1 \text{ or } 2$$



Assume: $IP3_{LNA}$=10 dBm, $IIP3_{MIX}$=6 dBm, G_{LNA}=15 dB, A_{f1}=0.2dB, A_{f2}=1 dB, L_c=10 dB
In linear units: $ip3_{LNA}$=10 mW, $iip3_{MIX}$=4 mw, g_{LNA}=31.6228, a_{f1}=1.0471, a_{f2}=1.2589

$IP3_{LNA}$ is at its output
→$IIP3_{LNA,dBm}$=$IP3_{LNA,dBm}$-$G_{LNA,dB}$
$IIP3_{MIX}$ is at its input

$$iip3_{rec} = \left(\left(\frac{g_{LNA}}{a_{f1} \cdot ip3_{LNA}}\right) + \left(\frac{g_{LNA}}{a_{f1} \cdot a_{f2} \cdot iip3_{MIX}}\right)\right)^{-1} =$$

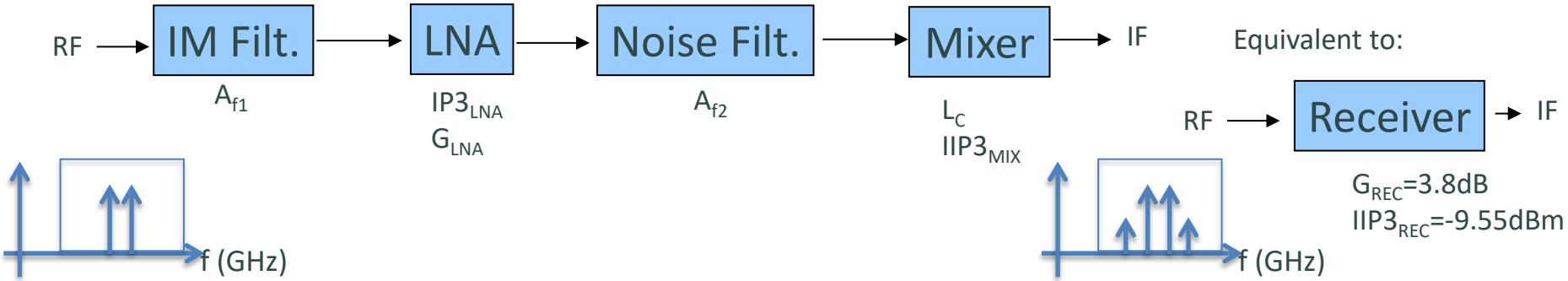
$$= \frac{1}{31.6228}\left(\frac{1}{1.047 \cdot 10} + \frac{1}{1.047 \cdot 1.2589 \cdot 4}\right)^{-1} = \frac{3.5065}{31.6228} = 0.1109 \text{ mW } (-9.55 \text{ dBm})$$

IIP3 improves (higher is better) by:

- improving IIP3s of the stages, especially of the last stages
- increasing filters attenuation in the passband (worse filters)
- decreasing the LNA gain

Not real solutions: the output signal power is lower…

Example of Equivalent IIP3 and Intermod. Power



The RF signal is 2 tones, of combined average power -30dBm → -33dBm each tone.

What is the spectrum at the mixer output (IF)?

The main tones at IF have power $P_{\omega 1,dBm}$=-33dBm+G_{REC}=-29.2dBm each

The closest 3rd order intermodulation products follow

$$P_{2\omega_2-\omega_{1dBm}} \approx 3P_{\omega 1_{dBm}} - 2IIP_{3_{dBm}} - 2G_{dB}$$

→$P_{2\omega 2\text{-}\omega 1dBm}$=3 (-29.2dBm)-2(-9.55dBm)-2 · 3.8dB=-76.1dBm

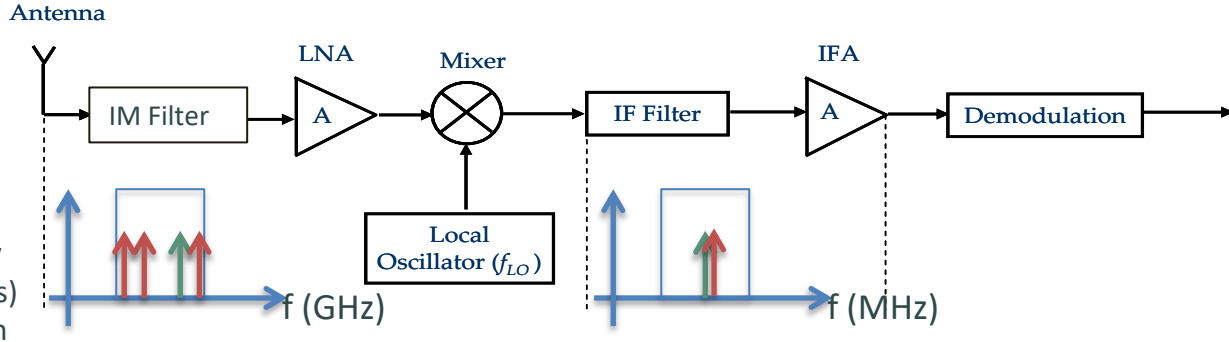
Spurious Signals in RX

Spurious: signals at different RF frequencies, possibly producing an output in the desired signal's base band

→overlapping and degrading the desired one

Spurious may be created:

- Externally → Often reduced/suppressed by filters (but difficult for broad band receivers)
- Internally → Typically by the mixer, through its intrinsic non linearity.
  Given IF and LO frequencies, some frequencies f_{RF} can be downconverted at same IF:

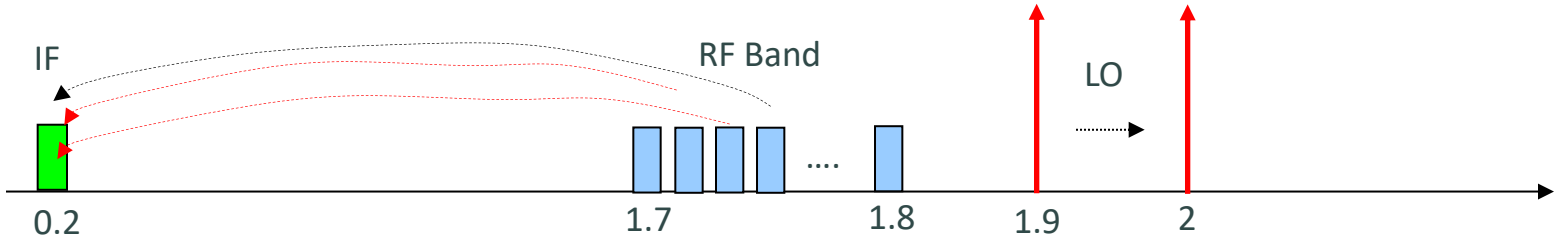


$$mf_{RF} - nf_{LO} = \pm f_{IF} \quad \Rightarrow \quad f_{RF} = \frac{nf_{LO} \pm f_{IF}}{m} \qquad n, m = 1,2,3\ldots$$

- With *m=n=1*: the desired and the image frequencies are obtained (IM band rejected by IM filter).
- All other values of m,n: predict some f_{RF} at which an undesired signal would be downconverted to the same f_{IF}.
- Spurious can be also generated when two high-power signals close to each other are present at the receiver input. The finite IIP3 of the receiver might create intermodulation products which superimpose to the useful signal

More General Approach to Spurious

Receiver band: 1700-1800 MHz (other bands are suppressed by the RF filter before the mixer)
Channel band: 1 MHz (101 channels)
IF frequency: 200 MHz
Local Oscillator:1900-2000 MHz
Image band (rejected): 2100-2200



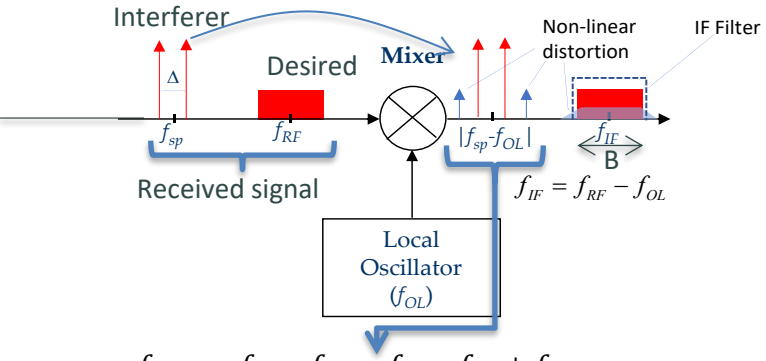
To receive channel 92 (f_{RF}=1.791GHz), we set LO to $200MHz=f_{LO}-f_{RF} \rightarrow f_{LO}=1.991GHz.$
Harmonics of LO are n · f_{LO} [GHz]: 3.982, 5.973…
Harmonics of the various channels will be at m · f_{RF}
Do any of these products fall into the 200MHz IF? $200MHz=n \cdot f_{LO} \pm m \cdot f_{RF} \rightarrow f_{RF}=(n \cdot 1.991GHz \pm 0.2)/m$

```
n=[1:8]; m=[1:8]; [nn,mm]=meshgrid(n,m); (nn.*1.991+0.2)./mm, (nn.*1.991-0.2)./mm
```

Three other frequencies in the RF band are converted at f_{IF} with the following values of m and n:
n=7, m=8: f_1=1.7171 (~channel 18), f_2=1.7671 (~channel 68); n=6, m=7: f_3=1.7351 (~channel 36)

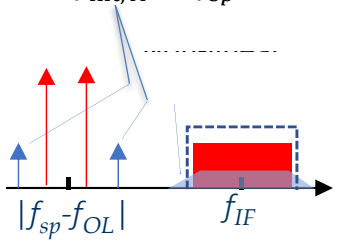
However: these spurious signals are from high order harmonics (6,7,8) → Level should be practically low

Another Phenomenon of Intermodulation in Downconversion



$$f_{sp,IF} = f_{sp} - f_{OL} = f_{sp} - f_{RF} + f_{IF}$$

$$f_{\text{int},IF} = f_{sp} \pm 1.5 \cdot \Delta - f_{OL}$$



$$f_{IF} = f_{RF} - f_{OL}$$

- Signals which are far from the RF band → removed by image filter
- Signals close to RF signal (within the image filter) which are converted OUTSIDE the IF band
  → removed by the IF channel filter.
- But the mixer has finite IIP3 → intermodulation products!

$$P_{\text{int},IF} = \left(3P_{sp} - 2IIP_3\right) - L_c$$

$P_{\text{int},IF}$: of each int. prod., dBm; P_{sp}: of each tone at input, dBm

L_c = conversion loss of the mixer

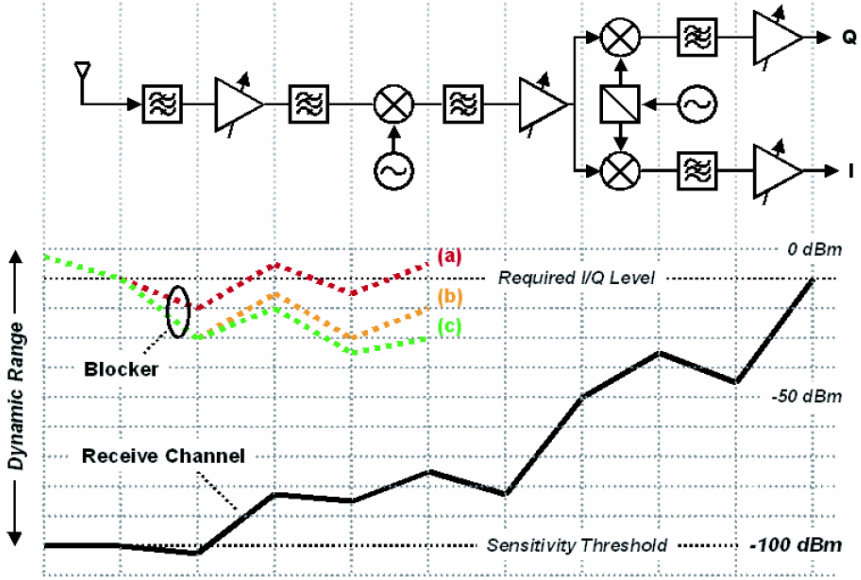
- Some intermodulation products by signals outside the RF band may fall INSIDE the IF! → Degradation/interference
- E.g., with two tones this occurs if (assuming $f_{sp} < f_{RF}$):

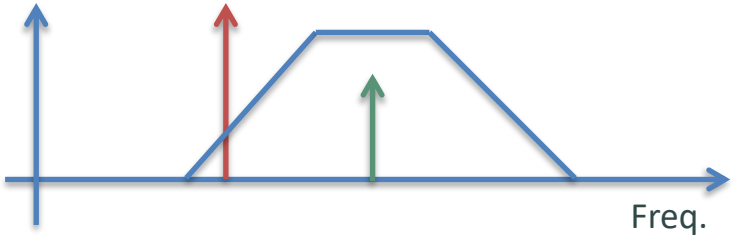
$$f_{\text{int},IF} > f_{IF} - \frac{B}{2} \Rightarrow f_{sp} + 1.5 \cdot \Delta - f_{OL} > f_{IF} - \frac{B}{2} \Rightarrow f_{sp} > f_{RF} - \frac{3 \cdot \Delta + B}{2}$$

$$f_{\text{int},IF} < f_{IF} + \frac{B}{2} \Rightarrow f_{sp} + 1.5 \cdot \Delta - f_{OL} < f_{IF} + \frac{B}{2} \Rightarrow f_{sp} < f_{RF} - \frac{3 \cdot \Delta - B}{2}$$

Blockers

When a receiver operates in a highly crowded frequency spectrum, it may happen that high power unwanted signals are outside the useful band but are not rejected enough by the image filter. Problem?



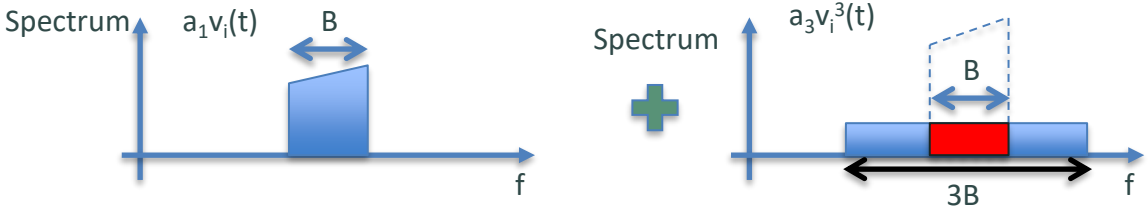


The disturbing signal may be very strong and, albeit mostly rejected by the RF filter, it might reach the LNA
→ Additional power enters the LNA
→ Further gain compression for the useful signal
→ Additional intermodulation distortion (IMD)

The blocker will be stopped later by the IF filter, but damages are already done.
More about blockers when discussing architectures…

Wrap-up: Non-linearity of a receiver

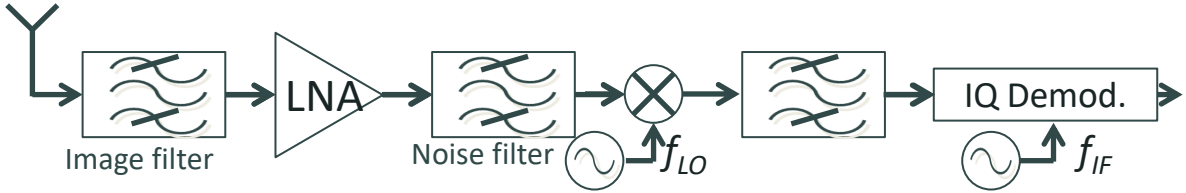
- Compression & intermodulation distortion (IMD) of the useful signal: example $a_1 v_i(t)+a_3 v_i^3(t)$



On a modulated signal, not only compression: also additional undesired signal is superimposed to the useful part
→Intermodulation distortion

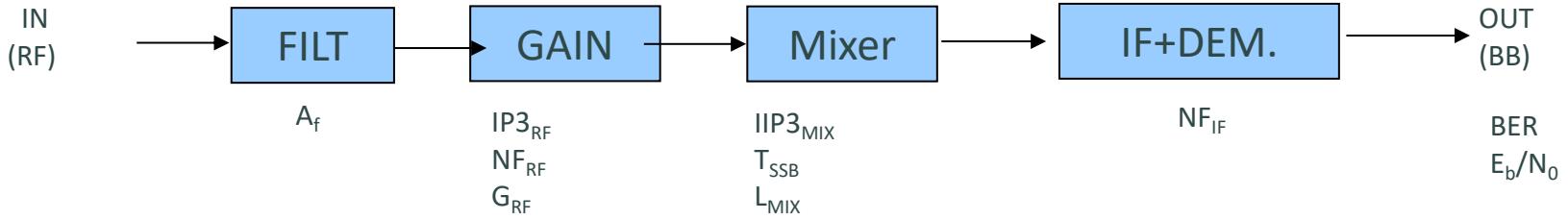
- Spurious signals: signals within the passband of the image filter (thus close to the useful one) which are downconverted to the same IF as the useful one due to harmonic mixing (*m*spurious signals - n*local oscillator* combinations which fall in the same IF), typically created by the strong non-linearity of the mixer
- Intermodulation products: signals within the passband of the image filter (thus close to the useful one) which intermodulate due to finite IIP3 of the whole receiver (LNA and/or mixer)
- Blockers: strong signals which are only partially attenuated by the image filter and can potentially "blind"the receiver (saturation) hence introducing more compression and intermodulation distortion on the useful signal

Receiver Budget Analysis



- Budget analysis: determine the individual specifications of the receiver blocks
- Bit error rate (BER): usually measures the quality of the received signal
- Sensitivity: lowest input RF signal that must be detected and returned by the receiver with the required BER
- Dynamic range: entire range of input RF power from the sensitivity threshold up to the maximum detected signal (determined by the non-linear distortion)
- The output of a receiver budget analysis: calculations for gain, noise figure, filtering, intermodulation products (IM), and 1dB compression power (at receiver input).
- CAD programs often offer specific tools for carrying out the budget analysis at system level

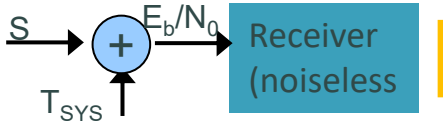
Receiver Budget Design (example)



Requirements (given):
- Sensitivity S (dBm) for a given BER at given modulation (→ E_b/N_0)
- Dynamic Range D (=$P_{max,dBm}$-S_{dBm})
- Data rate R (Mbit/sec) and signal bandwidth B (MHz)

<u>Constraint on the minimum input power</u>

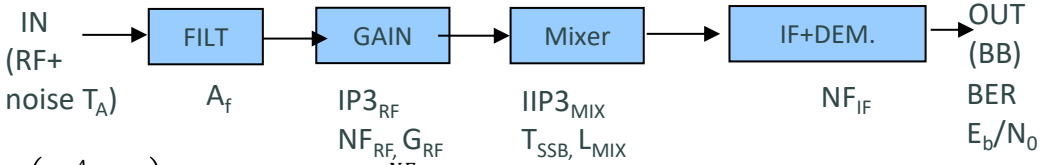
Scheme of the receiver with its equivalent system noise temperature (T_{SYS}) :



E_b/N_0 $\quad SNR = \frac{S}{(kT_{SYS}) \cdot B} = \frac{E_b}{N_0}\frac{R}{B}$

Compute T_{SYS}

Receiver Budget Evaluation (example)



T_{SYS} depends on temperatures and gains:

$$T_{SYS} = T_A + T_F + 2A_f T_{RF} + \frac{A_f(T_{SSB} + L_{mix}T_{IF})}{G_{RF}}; \quad T_F = 290K\left(10^{\frac{A_f}{10}} - 1\right); \quad T_{RF} = 290K\left(10^{\frac{NF_{RF}}{10}} - 1\right)$$

The budget equation is then $\frac{E_b}{N_0}\frac{R}{B} = \frac{S}{(KT_{SYS})\cdot B}$ which in dB is:

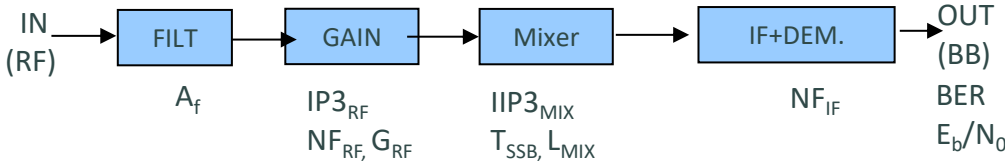
$$\left(\frac{E_b}{N_0}\right)_{dB} + \left(\frac{R}{B}\right)_{dB} = S\Big|_{dBm} - 10\log(KB) - 10\log\left(T_A + T_F + 2A_f T_{RF} + \frac{A_f(T_{SSB} + L_{MIX}T_{IF})}{G_{RF}}\right)$$

Given Given Given Given Var. Var. Var. Var. Var. Var.

- For the given requirements there are several combinations of the variables which satisfy the budget equation
- Selection criteria must include other possible constraints: costs, dimensions, availability of components etc.
- Some variables affect also the large signal behavior of the receiver (see next slide)

Receiver Budget Design (example)

$$\left(\frac{1}{iip3}\right)^n = \left(\frac{1}{iip3_1}\right)^n + \left(\frac{g_1}{iip3_2}\right)^n + \left(\frac{g_1 g_2}{iip3_3}\right)^n + \left(\frac{g_1 g_2 g_3}{iip3_4}\right)^n \quad n = 1 \text{ or } 2$$



Constraint on maximum input power

Sensitivity *S* and dynamic range *D* known

→ Find IIP3 via definition of Dynamic Range

DR_{dB}=$Pmax_{dBm}$-S_{dBm} (Pmax: input power giving equiv. input intermod. distortion power equal to *S*):

$$DR_{dB} = \frac{2}{3}\left(IIP_{3,dBm} + 3 - S_{dBm}\right)$$

IIP3 then depends on the receivers components through the formula of cascaded stages (remember that linear blocks have infinite IP3 but contribute through their gain or attenuation):

$$\left(\frac{1}{iip_3}\right)^2 = \left(\frac{\frac{1}{a_f}}{iip_{3,LNA}}\right)^2 + \left(\frac{\frac{g_{RF}}{a_f}}{iip_{3,mix}}\right)^2$$

Thus:

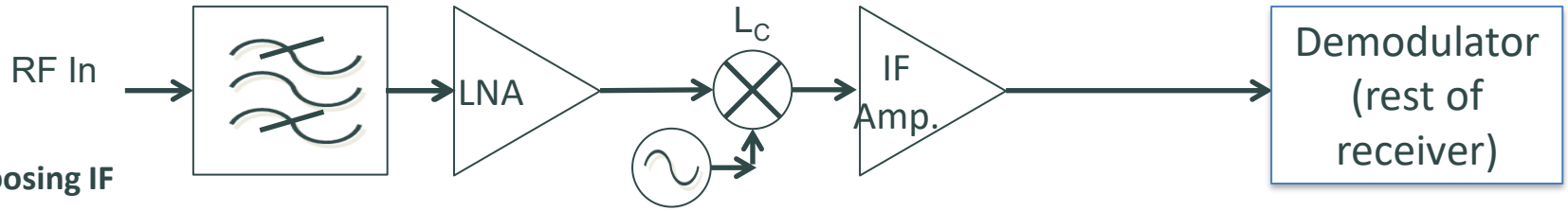
$$\left(\frac{\frac{1}{a_f}}{\frac{IP3_{RF}}{g_{RF}}}\right)^2 + \left(\frac{\frac{1}{a_f}g_{RF}}{IIP3_{mix}}\right)^2 = \left(\frac{1}{10^{\frac{\left(\frac{3}{2}\right)DR_{dB}-3+S_{dBm}}{10}}}\right)^2$$

Var. Var. Var. Var.

A_f and G_{RF} impact both on linear and non-linear equations

A real system design is tricky and involves several assumptions based on experience and tradeoffs!

Single Conversion Receiver → Double Conversion



Choosing IF

High IF: The image band is far from the desired band

→ Good image rejection even with a simple image filter (high rejection is provided) → Low cost

→ Can accommodate wideband RF signals

→ IF stages work at a high frequency (more expensive and difficult to design)

Low IF: The image band is close to the desired band

→ All IF stages work at low frequency, simpler to design, lower cost

→ Limits band of RF signals

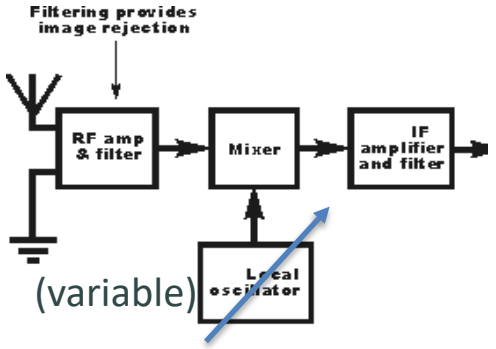
Conflicting requirements in Single-Conversion Receiver!

A possible solution for both requirements at the same time: double conversion architecture

Single vs Double Conversion Receiver

Single Conversion Receiver:

- It allows most of the signal amplification to occur at IF (easier than at RF, being f_{IF} fixed and much smaller than f_{RF})
- Channel filtering at IF (IF filters compact and not to be tuned)
- Flexibility in the choice of components (LNA, Mixer, Filters) allows to fit more easily the requirements (Gain, Noise, etc)



Double Conversion Receiver:

- A high intermediate frequency to achieve high levels of image rejection
- Then a low intermediate frequency to provide the levels of performance required for the adjacent channel selectivity and amplification
- More blocks/components/complexity

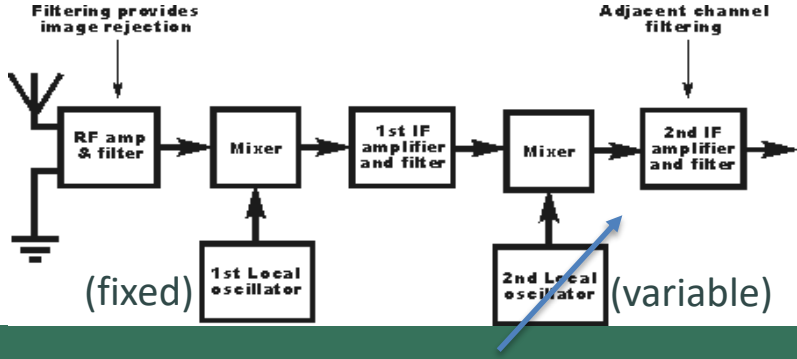
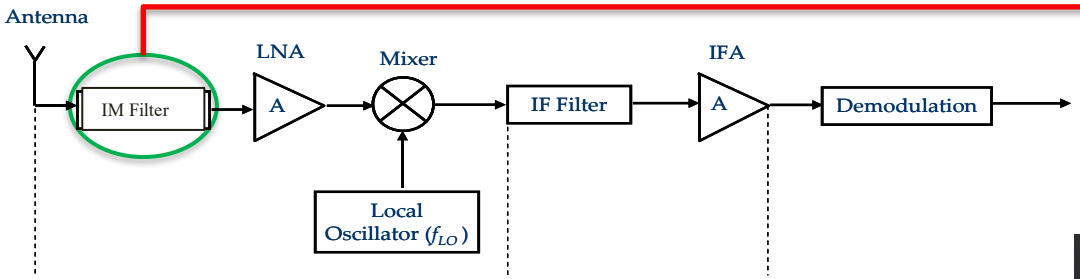
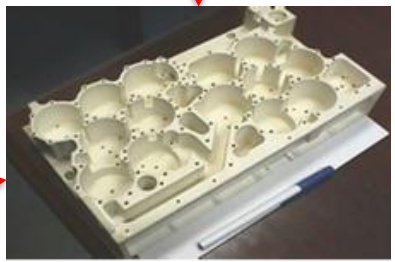


Image-reject receivers

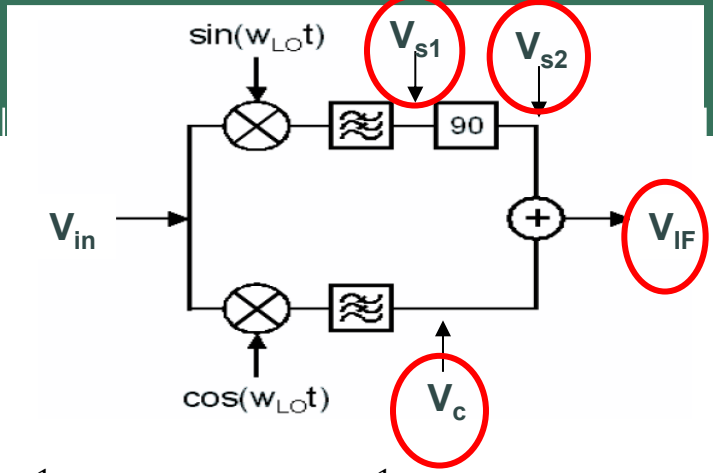


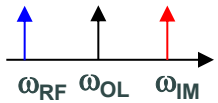


- Major drawbacks of heterodyne receiver: interference from image band
- Suppression of image band: requires filters with very high attenuation (60-90 dB), resulting in cumbersome devices (waveguides) expensive and not integrable at all with planar circuits
- Reduce suppression requirements: intrinsic suppression of the image frequency: image-reject mixer
- The receivers using such mixers (image-reject receivers) have relaxed requirements for the image filter (up to 35 dB less than classical receivers)

Cheaper, easier requirements, integration possible

Image-Reject Mixer





$$V_{in} = V_{RF}\cos(\omega_{RF}t) + V_{IM}\cos(\omega_{IM}t)$$

$$\cos\theta\cos\varphi = \tfrac{1}{2}(\cos(\theta-\varphi)+\cos(\theta+\varphi))$$
$$\sin\theta\sin\varphi = \tfrac{1}{2}(\cos(\theta-\varphi)-\cos(\theta+\varphi))$$
$$\sin\theta\cos\varphi = \tfrac{1}{2}(\sin(\theta+\varphi)+\sin(\theta-\varphi))$$
$$\cos\theta\sin\varphi = \tfrac{1}{2}(\sin(\theta+\varphi)-\sin(\theta-\varphi))$$

$$V_{s1} = V_{RF}\cos(\omega_{RF}t)\sin(\omega_{LO}t) + V_{IM}\cos(\omega_{IM}t)\sin(\omega_{LO}t) = \frac{1}{2}V_{RF}\sin\big((\omega_{LO}-\omega_{RF})t\big) - \frac{1}{2}V_{IM}\sin\big((\omega_{IM}-\omega_{LO})t\big)$$

$$V_{s2} = \frac{1}{2}V_{RF}\cos\big((\omega_{LO}-\omega_{RF})t\big) - \frac{1}{2}V_{IM}\cos\big((\omega_{IM}-\omega_{LO})t\big)$$

$$\sin\left(x+\frac{\pi}{2}\right) = \cos(x)$$

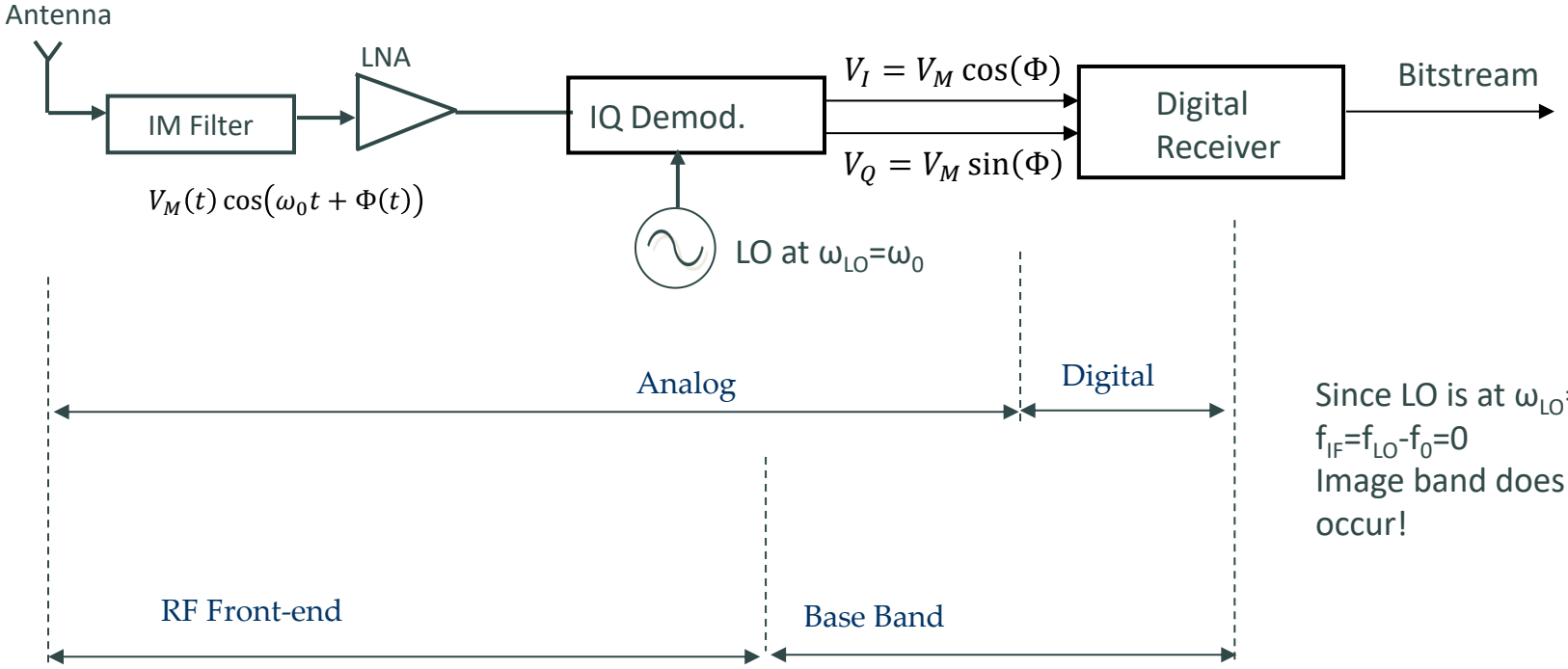
$$V_c = V_{RF}\cos(\omega_{RF}t)\cos(\omega_{OL}t) + V_{IM}\cos(\omega_{IM}t)\cos(\omega_{OL}t) = \frac{1}{2}V_{RF}\cos\big((\omega_{OL}-\omega_{RF})t\big) + \frac{1}{2}V_{IM}\cos\big((\omega_{IM}-\omega_{OL})t\big) + \cdots$$

$$V_{IF} = V_c + V_{s2} = \frac{1}{2}V_{RF}\cos\big((\omega_{OL}-\omega_{RF})t\big) - \cancel{\frac{1}{2}V_{IM}\cos\big((\omega_{IM}-\omega_{OL})t\big)} + \frac{1}{2}V_{RF}\cos\big((\omega_{OL}-\omega_{RF})t\big) + \cancel{\frac{1}{2}V_{IM}\cos\big((\omega_{IM}-\omega_{OL})t\big)} =$$

$$V_{IF} = V_{RF}\cos\big((\omega_{OL}-\omega_{RF})t\big)$$

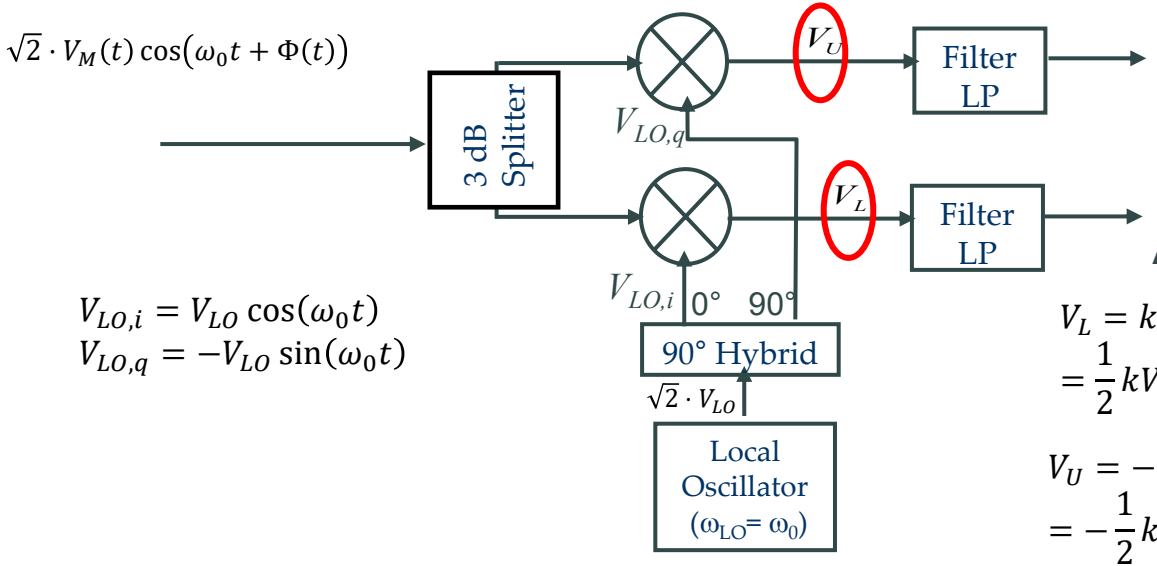
- Perfect suppression of image (ideally)
- More than double hardware than a standard mixer

Direct-conversion Receiver (zero IF)



Since LO is at $\omega_{LO}=\omega_0$:
$f_{IF}=f_{LO}-f_0=0$
Image band does not occur!

IQ Demodulator (RF signal directly converted to baseband)



At the mixers out:

$$V_{LO,i} = V_{LO} \cos(\omega_0 t)$$
$$V_{LO,q} = -V_{LO} \sin(\omega_0 t)$$

$$V_L = kV_{LO}V_M \cos(\omega_0 t + \Phi) \cos(\omega_0 t) =$$
$$= \frac{1}{2} kV_{LO}V_M (\cos(2\omega_0 t + \Phi) + \cos(\Phi))$$

$$V_U = -kV_{LO}V_M \cos(\omega_0 t + \Phi) \sin(\omega_0 t) =$$
$$= -\frac{1}{2} kV_{LO}V_M (\cos(2\omega_0 t + \Phi) + \sin(\Phi))$$

The LP (low pass) filters remove the components at $2\omega_0$ (and all the others produced by the actual nonlinearities). The two output signals are then proportional to I and Q components of the modulated RF signal

The IQ demod. is also used as final downconversion after IF in Single and Double-Conversion receivers (thus it can be implemented in the digital domain if the IF is low enough to be sampled by the ADC)

Pro & Cons of direct-conversion receiver

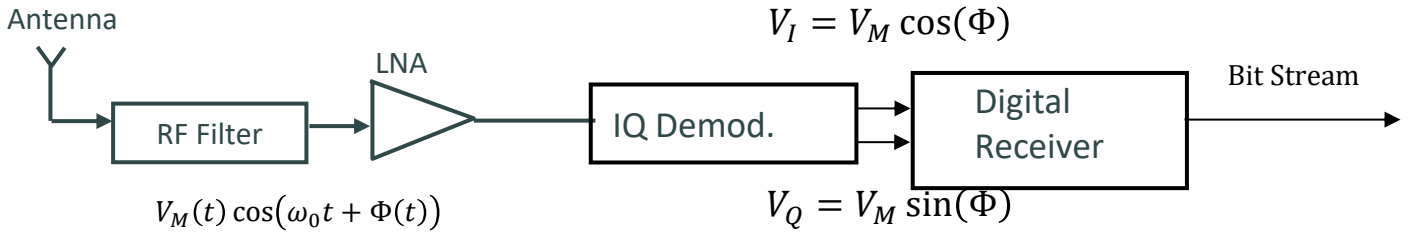
Reduced complexity (cheaper and more compact)

Easier to integrate (but not the RF Filter)

More susceptible to noise and distortion

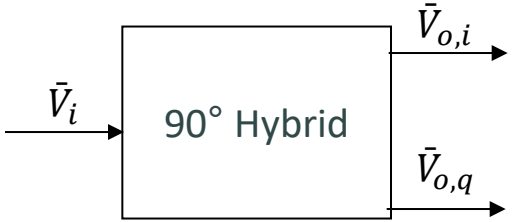
DC offset voltage at the mixer output may be present (since the RF signal might also contains a useful component at f_0)

Offset voltage at the mixer output not easy to eliminate (it overlaps with the useful signal)



Hybrid Quadrature Splitter

The hybrid splitter is a passive 3 port device which divides the input in two copies and adds a phase (assuming all ports terminated on the reference load R_0):



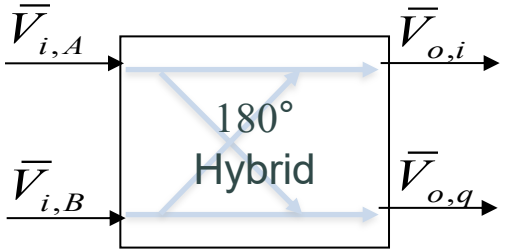
$$\bar{V}_{o,i} = \frac{1}{\sqrt{2}}\bar{V}_i, \qquad \bar{V}_{o,q} = -j\frac{1}{\sqrt{2}}\bar{V}_i$$

All the signals are voltage phasors (carrier f_0)

The input signal is split into 2 equal signals carrying half of the overall power. → -3dB each
The two output signals have a phase difference of 90°.
All the ports are matched (no reflected power).
Splitters with 0° and 180° of phase difference are also possible.

Hybrid Combiner

A passive 4-port device characterized by (all the ports are connected to the reference load R_0):



$$\bar{V}_{o,i} = \frac{1}{\sqrt{2}}\left(\bar{V}_{i,A} + \bar{V}_{i,B}\right)$$
$$\bar{V}_{o,q} = \frac{1}{\sqrt{2}}\left(\bar{V}_{i,A} - \bar{V}_{i,B}\right)$$

All the signals are voltage phasors (at carrier f_0)

When the signals are equal, $V_{o,i}$=sqrt(2)$V_{i,A}$ and $V_{o,q}$=0.
When one of the inputs is 0, the other is simply split (with 0 and 180° phase): works also as 180° hybrid splitter!

When the signal are statistically defined:

$$P_{o,i} = \frac{1}{2}\frac{\left|\bar{V}_{o,i}\right|^2}{R_0} = \frac{1}{2}\left(\frac{\left|\bar{V}_{i,A} + \bar{V}_{i,B}\right|^2}{2R_0}\right) = \frac{1}{2}\left(\frac{\left|\bar{V}_{i,A}\right|^2}{2R_0} + \frac{\left|\bar{V}_{i,B}\right|^2}{2R_0} + \frac{\langle \bar{V}_{i,A} \times \bar{V}_{i,B} \rangle}{R_0}\right) = \frac{1}{2}\left(P_{i,A} + P_{i,B} + P_{cross}\right)$$

For uncorrelated signals P_{cross}=0, then:

$$P_{o,i} = \frac{1}{2}\left(P_{i,A} + P_{i,B}\right)$$

The output power is 3dB less than the sum of the input powers